

ROBOTVISION.M

```
*****
% Red Ball Tracker - Robot Vision
% Software used Matlab 7.0.1(R14)
% Algorithm used RGB Colour Detection
*****
% SOME LINES HERE ARE COMMENTED OUT FOR
% FASTER PROCESSING. THE READER SHOULD UN-
% COMMENT THEM WHEN NEEDED

clear;
clc % Clearing Matlab desktop
vid=videoinput('winvideo',1,'RGB24_
640X480'); % Defining the video input
object
set(vid,'FrameRateTrigger',1); % Setting
frames per trigger
% preview(vid); %///// Showing the
video of the moving Ball (TO BE USED %
% WHILE TESTING)

pause(10); % Waiting for a certain time
for the system to get initialised
rgb_image = getsnapshot(vid); % Storing
Image in a global variable
[a b c] = size(rgb_image); % Determining
the size of the captured frame.
y=a;
x=b;
% Defining Boundaries
x1=x/2-120;
x2=x/2+120;
y1=y/2-30;
y2=y/2+30;
ser=serial('COM1'); % Defining the speci-
fied COM Port to be used
fopen(ser); % Starting serial
```

```
Communication, opening serial port
while(1)
    rgb_image = getsnapshot(vid); %
    storing image in a global variable
    flushdata(vid); % Flushing the buffer
    rbar=0;
    cbar=0;
    e=0;

    fR=rgb_image(:,1); fG=rgb_
    image(:,2); fB=rgb_image(:,3); % Storing
    RGB components of the image in
    separate arrays
    I=((fR>140) & (fG<105) &
    (fB<100)); % Converting the RGB Image
    into binary image // Detecting only the
    red component
    % Following are the steps for De-
    tecting the red ball
    se=strel('disk',20);
    B=imopen(I,se);
    final=imclose(B,se);
    [L,n]=bwlabel(final);
    % imshow(rgb_image); %///// THIS IS TO
    BE USED ONLY WHILE TESTING
    hold on % ///// THIS IS TO BE USED
    ONLY WHILE TESTING
    for k=1:n
        [r,c]=find(L==k);
        rbar=mean(r);
        cbar=mean(c);
        % plot(cbar,rbar,'Marker','s','
        MarkerEdgeColor','b','MarkerSize',20)
        %///// THIS IS TO BE USED ONLY WHILE TEST-
        ING
        e=((cbar<x1)*2+2)+
        ((cbar<x2)*2)+ ((rbar<y1)*2)+
```

```
(rbar<y2)) % Converting to decimal
number
    end
    % Decision Making Conditions
    switch (e)
        case 5
            disp('Move
            left'), fprintf(ser,'L');
            case 6
            disp('Move
            left'), fprintf(ser,'L');
            case 7
            disp('Move
            left'), fprintf(ser,'L');
            case 9
            disp('Move right'), fprintf
            (ser,'R');
            case 10
            disp('Move right'), fprintf
            (ser,'R');
            case 11
            disp('Move right'), fprintf
            (ser,'R');
            case 13
            disp('Move forward'), fprin
            tf(ser,'F');
            case 14
            disp('Move
            back'), fprintf(ser,'B');
            otherwise
            disp('Stop Moving'), fprin
            tf(ser,'S');
        end
    end
    fclose(ser); % closing serial port
```

Do it Yourself Electronic Kits and learn in the process

Choose from a wide range of kits to support your creative ventures. Kits 'n' Spares, a venture of Efy Group, brings to you the kits, complete with components, circuits, PCBs, hard-to-find ICs and easy-to-use instruction manuals

NEW KITS

- Chip-Talk Hobby Projects
- Cell Phone Operated Land Over
- Micro-Ultrasonic Distance Meter
- Secured Room Access System
- Microcontroller-based Bi-directional
- Microcontroller Based Code Lock
- Microcontroller Based Heart Rate Meter
- Microcontroller Based Tachometer
- Automatic Low Power Emergency Light
- PC-based Automatic Time Manager
- FM Booster

Rs 1990/-
Rs 2500/-
Rs 1150/-
Rs 1000/-
Rs 900/-
Rs 800/-
Rs 650/-
Rs 500/-
Rs 450/-
Rs 250/-
Rs 60/-

* Please add 4% tax for local sale in delhi or 4% for order outside Delhi.
Forwarding charges Rs 40/-per kit.



For the 'engineer in you'

Kits'n'Spares

Please send your remittance by DD/MO* to :

Kits'n'Spares

D-8/5, Okhla Industrial Area, Phase-1,
New Delhi 110020; Phone: 26371661, 26371662
E-Mail: kits@efyindia.com, kitsmktg@efyindia.com
Website: www.kitsnspares.com

Also available at Kits'n'Spares Outlet:

EFY Tech Center

303, Dohil Chambers, 46, Nehru Place,
New Delhi-110019; Phone: 46597697

To view our entire range of products and to purchase them, visit our website: www.kitsnspares.com

OUR DEALER NETWORK

■ AHMEDABAD—RELIEF ROAD. Ph: 09825200451, 079-22139841; NARANPURA—Ph: 9898121289 ■ AIZWAL—Ph: 0389-2310719, 2311909, 09837055376 ■ AMBALA—Ph: 09315387122, 0941611330 ■ ASSAM(GUWAHATI)—Ph: 09357193093 ■ BANGLORE—Ph: 0801446473, 09341268861 ■ BOPAL—Ph: 0755-4281576, 0926554365 ■ BHUJANESHWAR—Ph: 09437279441 ■ BHATINDA—Ph: 09417223151 ■ BHILAI NAGAR—Ph: 0982192818 ■ BIRAILLY—Ph: 0581-2556675, 09837055376 ■ CALCUTTA—Ph: 033-4947123, 32544558, 09433012210 ■ CHANDIGARH—Ph: 09815140504 ■ CHENNAI—Ph: 044-42046343, 09994545727 ■ CUDDALORE—Ph: 09344412298 ■ COIMBATORE—Ph: 0422-2394113, 2381083 ■ DELHI-JANAKPURI—Ph: 09899795966, 22501716 ■ DEHRADUN—Ph: 0412369744 ■ GHAZIABAD—BOOK WORLD, Ph: 09810638616; ORBITECH—Ph: 9891661368 ■ HISAR—Ph: 09466662123, 09812260340 ■ HYDERABAD—Ph: 040-24756823, 09849070411 ■ JALANDHAR—Ph: 09988299442, 0181-2602314 ■ JAIPUR—Ph: 09828123992 ■ JODHPUR—Ph: 09828123958 ■ KOLKATA—Ph: 033-2546558, 24947123 ■ LUDHIANA—Ph: 09815742497 ■ LUCKNOW—Ph: 09454109057 ■ MADURAI—Ph: 09842103477, 09842119512 ■ MANGLORE—Ph: 0824-2429533 ■ MUMBAI—Ph: 09867478289 ■ NAGARCOIL—Ph: 04652-222505 ■ NAGALAND—Ph: 03682-229799, 09436006160 ■ Noida—Ph: 9212331453 ■ ODISHA—Ph: 09443117715 ■ PUNE—Ph: 020-24300597, 09422555754 ■ SOLAN—Ph: 09816076344 ■ SECUNDRABAD—Ph: 09396699217 ■ SHOLAPUR—Ph: 0217-2720099, 09823383081 ■ THIRISSUR—Ph: 0487-2427567, 2427629 ■ TRICHY—Ph: 0431-2791294 ■ VARANASI—Ph: 0542-2277357